

Exploratory Testing Notes – Tic-Tac-Toe

Objective

Explore the Tic-Tac-Toe application to understand its functionality, identify potential defects, and assess the overall user experience.

Areas Explored

- Game board interactions
- Difficulty level selection
- Computer AI behavior
- Hint functionality
- New Game and Reset actions
- Game state transitions

Observations & Potential Defects

DEFECT-01: Difficulty Levels Do Not Appear to Affect Gameplay

Observation:

After playing multiple games using different difficulty levels, the computer's behavior appeared identical across all levels.

Expected:

Different difficulty levels should noticeably affect the AI's decision-making and challenge level.

Actual:

No observable difference was detected between difficulty levels.

Severity: Medium

DEFECT-02: Computer Selects an Already Occupied Cell

Observation:

In some game sessions, after selecting a cell, the computer appeared to place its move in the same cell that was already occupied.

Expected:

The computer should only select empty cells.

Actual:

The computer occasionally attempts to use an occupied cell.

Severity: High

DEFECT-03: Predictable AI Behavior

Observation:

The computer frequently follows the same move patterns and strategies across multiple games.

Expected:

The AI should demonstrate more adaptive or varied behavior, especially in a game intended to simulate an opponent.

Actual:

Gameplay becomes predictable due to repetitive decision-making.

Severity: Medium

DEFECT-04: New Game and Reset Produce the Same Result

Observation:

The "New Game" and "Reset" buttons appear to perform the same action.

Expected:

Each button should have a clearly differentiated purpose, or the UI should avoid redundant functionality.

Actual:

No noticeable difference in behavior was observed.

Severity: Low

DEFECT-05: Hint Functionality Provides Incorrect Recommendations

Observation:

The "Get Hint" feature occasionally suggests moves that are strategically poor or fail to block an immediate winning opportunity.

Expected:

Hints should recommend the optimal or near-optimal move for the current game state.

Actual:

Some recommendations appear incorrect or suboptimal.

Severity: High

Risks Identified

- AI functionality may not be fully implemented.
- Difficulty settings may not be connected to game logic.
- Hint algorithm may not use the same logic as the game AI.
- Redundant controls may confuse users.

Recommendation

Perform deeper functional testing around AI decision-making, game-state management, and hint generation logic before release.